

The Latest In SCOPES



Abhimanyu Rathore is a content specialist at EBY

Although the performance of oscilloscopes has consistently improved over the years, their basic functionality remains the same. "Whether the application is monitoring a patient's heart signal or debugging the timing glitches at the output of a field-programmable gate array, in general, these display the voltage variation of the signals connected to their input channels with respect to time," explains Shayan Ushani, editor at Analog Arts Inc.

There is never enough bandwidth

Hardware changes at the analogue-to-digital converter (ADC) side have helped deliver more resolution to oscilloscopes and delivered 10-bit resolution where these previously used to max out at eight bits.

"We now see 10-bit ADCs with up to 8GHz of analogue bandwidth. Previously, 8-bit ADCs only had up to 2GHz of bandwidth at a full sampling rate of 20 gigasamples per second," says Akash Srivastava, application engineer, Keysight Technologies. Embedded scopes that have a higher waveform update have also seen an increase from 2GHz to 6GHz of bandwidth.

Points to note

- While digital oscilloscopes are dominating the market now, analogue oscilloscopes still play an important role for real-time signals and waveform displays.
- Requirement for mixed-signal oscilloscopes began in early 2000s, but nowadays engineers need a mixed-domain oscilloscope to carry out frequency and time-domain analysis simultaneously.
- Phone and wristwatch based oscilloscope have emerged as good alternatives in certain applications.
- Movement from knobs to touchscreens has happened, but model acceptance depends on users' taste for experience.

Who needs this kind of bandwidth? Srivastava says that, in today's world, 8GHz is almost a general-purpose requirement for engineers working on gigabit Ethernet or universal serial bus (USB 2.0).

Embedded-system oscilloscopes have a signature specification called waveform update rate, which helps in reducing the dead time between captured data. "These are general-purpose but are specifically used by engineers working on capturing intermittent signals that require a fast update rate. Detecting a glitch that just happens once in a while requires a higher waveform rate and higher bandwidth," explains Srivastava.

Adding intelligence to the scope

New PCI eXtensions for Instrumentation (PXI) oscilloscopes come with programmable modules. "Normally, all scopes that we use acquire all data but show only an insignificant part of that data with the result that, what you see on the liquid crystal display need not be what you are acquiring. By using field-programmable gate arrays, you can acquire data without losing a single sample, and this makes great sense in applications like astrophysics and medical images where you need continuous logging," explains Raviteja

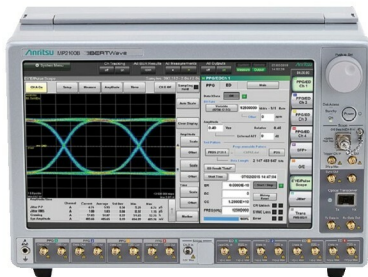


Fig. 1: MP2100B front top